

Algebra First-Year Exam, August 2021, Part 1

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let G be a cyclic group. Prove that every subgroup of G is cyclic. (Be sure to consider infinite cyclic groups as well as finite ones.)
2. Let G, H be finite groups with orders m, n .

(a) Prove that if m and n are coprime then

$$\text{Aut}(G \times H) \cong \text{Aut}(G) \times \text{Aut}(H).$$

(b) Give an example where m and n are not relatively prime and

$$\text{Aut}(G \times H) \not\cong \text{Aut}(G) \times \text{Aut}(H).$$

Justify your claims.

3. This problem has two parts.
 - (a) Let G be a finite group of order n and let p be the smallest prime which divides n . Prove that if H is a subgroup of G of index p then H is normal in G .
 - (b) Give an example of a finite group G and a subgroup $H \leq G$ such that H is not normal in G and the index $|G : H|$ is prime.
4. Let p, q, r be primes such that $p < q < r$ and let G be a group of order pqr . Prove that G has a normal Sylow subgroup.
5. Prove that the alternating group A_5 is simple. Prove that A_4 is solvable.
6. Show that there are four different homomorphisms $\phi : \mathbb{Z}_2 \rightarrow \text{Aut}(\mathbb{Z}_8)$, and prove that the corresponding semidirect products $\mathbb{Z}_8 \rtimes_{\phi} \mathbb{Z}_2$ are pairwise nonisomorphic. (Hint: Consider the number of elements of order 2.)